2018 H2 Physics Prelim P3 Suggested Solution

1(a) Gravitational potential energy of an object is the energy it possesses by virtue of its position in a gravitational field while elastic potential energy is the energy that is stored when a spring is stretched or compressed.

(b)(i) Taking the child and the pogo stick as a system.

At **A**, the gravitational potential energy is at a minimum while the elastic potential energy is at a maximum, since it is at rest, kinetic energy is zero.
From **A** to **B**, the gravitational potential energy has increased while the elastic potential energy is now zero since the spring is relaxed. Kinetic energy is non-zero at this point.
From **B** to **C**, the gravitational potential energy reaches its maximum, elastic potential energy is still zero while the kinetic energy is zero since it comes to rest.

(b)(ii) By conservation of energy,

Total energy at point A = Total energy at point C

$$\frac{1}{2}k{x_A}^2 = mg(x_A + x_C)$$